

MANASA TOPUGONDA

DATA ANALYST

Overland Park, Kansas | manasatopugonda123@gmail.com | +1 913-709-3684 | [LinkedIn](#)

SUMMARY

Accomplished **Data Analyst** with **4+ years of experience** across healthcare, technology, and finance domains, leveraging **SQL**, **Python**, and **R** to extract, analyze, and visualize large-scale structured and semi-structured datasets. Proficient in developing interactive dashboards using **Power BI**, **Tableau**, and **Excel** to support real-time performance monitoring, **KPI tracking**, and executive decision-making. Skilled in building and automating **ETL pipelines** with **AWS Glue**, **Azure Data Factory**, **SSIS**, and **Snowflake**, improving data reliability, reducing latency, and minimizing manual processing. Experienced in applying machine learning techniques such as regression, clustering, decision trees, and **NLP** using Scikit-learn, **TensorFlow**, and **PySpark** for predictive analytics and risk modeling. Strong background in data warehousing, **data governance**, and cloud-based analytics with **Redshift**, **PostgreSQL**, **MongoDB**, and **Apache Spark**, ensuring scalable, compliant, and high-quality data solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, R
- **Data Visualization:** Power BI, Tableau, MS Excel (VLOOKUP, Pivot Tables, Macros, VBA)
- **Big Data & Databases:** MySQL, MongoDB, PostgreSQL, BigQuery, AWS Redshift, SQL Server, SSIS, SSAS, ETL
- **Cloud Technologies:** AWS (S3, EC2, Lambda, Glue), Azure
- **Data Processing Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Apache Spark, PySpark
- **Machine Learning:** Regression, Classification, Clustering (K-Means, DBSCAN), Decision Trees, Neural Networks, NLP
- **Data Warehousing & Modeling:** Snowflake, Data Warehousing, Data Governance, Data Cleaning, Data Wrangling, Data Modeling, Predictive Analytics, Data Mining, Automation Scripts
- **Methodologies:** SDLC, Agile, Scrum

PROFESSIONAL EXPERIENCES

Blue Cross and Blue Shield, Kansas

Feb 2025 – Present

Data Analyst

- Designed **interactive dashboards in Power BI** to track provider performance, claims status, and patient risk metrics, reducing reporting time by **6 hours/week**.
- Built **predictive models** using **Scikit-learn** and **logistic regression** to identify high-risk patient populations, contributing to targeted care programs covering **25,000+ patients**.
- Automated **ETL pipelines** using **AWS Glue** and **SQL Server**, decreasing manual data handling errors by **40%** across clinical data systems.
- Conducted **anomaly detection** using **Python (Pandas, NumPy)** on pharmacy claims data, identifying **700+ potential billing discrepancies** per month.
- Implemented **HIPAA-compliant data governance protocols**, improving audit readiness and ensuring **95% compliance** with PHI data policies.
- Developed **custom SQL queries** on **Redshift** for ad-hoc financial analytics, shortening decision turnaround time by **22%**.
- Collaborated with data engineering teams in **Agile sprints**, enhancing backlog delivery velocity by **15%** using **Jira** and **Scrum** ceremonies.

Accenture, Hyderabad

Oct 2021 - Jul 2023

Data Analyst

- Developed **end-to-end ETL pipelines** using **SSIS**, **Azure Data Factory**, and **AWS Glue**, integrating data from ERP, CRM, and external APIs, improving data reliability and reducing processing latency by **30%**.
- Designed and deployed **interactive Power BI and Tableau dashboards** to track operational KPIs, enabling leadership to make **real-time business decisions**.
- Automated large-scale data processing using **Python (Pandas, NumPy)** and **PySpark**, reducing manual effort by **15+ hours/week**.
- Applied **machine learning models** (regression, clustering, NLP) in Python to generate actionable insights from customer and financial datasets, driving **\$40K in annual savings**.
- Built and maintained **data warehouses** in **Snowflake** and **SQL Server**, implementing **data governance and quality frameworks** to reduce errors by **25%**.
- Partnered with cross-functional stakeholders to **translate business requirements into BI and predictive analytics solutions**, ensuring strategic alignment.
- Mentored junior analysts on **SQL optimization**, **Python automation**, and **dashboard development**, improving overall team efficiency.

Dell Technologies, Hyderabad, Ind

Aug 2020 - Sep 2021

Data Analyst

- Designed and optimized **SQL queries** on **PostgreSQL**, **SQL Server**, and **MySQL** handling **10M+ records**, reducing report generation time by **40%**.

- Created **Power BI/Tableau dashboards** for sales, logistics, and finance teams, providing executives with data-driven insights that led to **\$60K in process savings**.
- Engineered **predictive models** (classification, decision trees, anomaly detection) using **Python, Scikit-learn, and R** to forecast order fulfillment risks and improve customer satisfaction.
- Automated recurring reporting tasks with **Excel Macros/VBA and Python scripts**, reducing manual work by **20 hours/month**.
- Conducted **data wrangling and cleaning** using **PySpark and Pandas** to prepare structured datasets for reporting and ML pipelines.
- Implemented **data warehouse schemas (Star & Snowflake)** and standardized KPI definitions, ensuring consistency across global reporting systems.
- Collaborated in **Agile Scrum teams** using Jira, contributing to sprint planning, backlog refinement, and delivering analytics features in line with business goals.

EDUCATION

- | | |
|--|----------------------------|
| • Master’s in Computer Information Systems
University of Central Missouri, Missouri, USA | Aug 2023 - Dec 2024 |
| • Bachelor’s in Computer Science and Engineering
CMR Technical Campus, Hyderabad, India | Jun 2017 - Sep 2021 |

PROJECTS

1. Epidemic Outbreak Prediction System

Technologies: Python, Pandas, Scikit-learn, Time Series Forecasting, Jupyter Notebooks, SQL, Matplotlib

- Developed a machine learning model to predict regional epidemic outbreaks using historical and real-time health data.
- Applied regression and time series forecasting to identify disease trends and improve emergency response planning.
- Built an alert system integrated with public health dashboards to provide early warnings to healthcare authorities.

2. Cryptocurrency Price Trend Analyzer

Technologies: Python, NumPy, Pandas, Plotly, Flask, XGBoost, LSTM, SpaCy, Sentiment Analysis

- Designed a prediction engine to forecast cryptocurrency price movements using historical data and social media sentiment.
- Integrated NLP models to analyze real-time sentiment trends, enhancing predictive accuracy.
- Built a web-based dashboard with Plotly and Flask for visualizing crypto price trends and volatility metrics.

3. Real-Time Fraud Detection in Financial Transactions

Technologies: Apache Kafka, Spark Streaming, HDFS, Scala, Cassandra, Power BI

- Engineered a real-time fraud detection pipeline using Kafka and Spark Streaming for high-throughput transaction analysis.
- Implemented anomaly detection using rule-based logic and machine learning scoring, reducing fraud risk by 40%.
- Visualized transaction heatmaps and key fraud metrics using Power BI dashboards for business and compliance teams.